

Major Natural resources of Nepal

Water Resources:

Despite being landlocked and lacking access to maritime facilities, Nepal is considered the second most water-rich country in the world after Brazil and the first in Asia. Nepal has over 6,000 rivers and streams. Among them, the three major rivers—Koshi, Gandaki, and Karnali—account for a water storage capacity of 148,000 cubic meters, which is 74% of the total water storage capacity.

Nepal's total hydropower production potential is approximately 83,000 megawatts. Based on the flow, water storage, and utility aspects of its rivers, Nepal's rivers can be categorized into three classifications as follows:

(a) **First-Level Rivers:** Rivers that originate from the melting snow of the Himalayas or from the Himalayan region and flow continuously with large volumes of water, such as the Sapta Koshi, Sapta Gandaki, and Karnali, are categorized as first-level rivers. These rivers are considered suitable for large hydropower and irrigation projects.

(b) **Second-Level Rivers:** Rivers that originate from the Mahabharat mountain range, located below the Himalayas, and flow continuously without drying up are classified as second-level rivers. However, these rivers experience significant variations in water volume, with higher flow during the monsoon and reduced flow in winter. Examples include Mechi, Kankai, Triyuga, Kamala, Bagmati, Banganga, Tinau, Rapti, Babai, and Mohana.

(c) **Third-Level Rivers:** Rivers that originate from the Siwalik mountain range (Chure hills) and exhibit a temporary nature, swelling significantly during the monsoon and reducing drastically or drying up during the winter, are classified as third-level rivers. Examples include Sirsiya, Tilabe, Jamuni, Hardinath, Dunduwa, and Arjun Khola. These rivers are primarily useful for rain-fed agriculture and short-term irrigation purposes.

Rivers of Nepal:

Nepal has over 6,000 rivers and streams, with a total length exceeding 45,000 kilometers. Due to the flow capacity of rivers originating from the Himalayan region and flowing continuously, the country has a hydropower potential of more than 83,000 megawatts. However, from an economic and technical perspective, it is estimated that 42,000 megawatts of hydropower can be practically harnessed.

Rivers and their capacity of producing Hydroelectricity:

The three major rivers of Nepal—Koshi, Gandaki, and Karnali—account for 90% of the total water resources, with significant hydropower production potential.

Karnali River: 32,000 MW

Koshi River: 22,000 MW

Gandaki River: 21,000 MW

Other Rivers: 8,000 MW

Rivers originating from Tibet:

1. Arun
2. Tamakoshi
3. Karnali
4. Kishuli
5. Sun Koshi
6. Timor

By what name are the rivers of Nepal known after they reach India?

Kankai – Wiring

Koshi – Damodar

Mechi – Mahananda

Mahakali – Sharada

Babai – Sarayu

Karnali – Ghaghara

Place of Origin of Rivers:

Saptakoshi-

Tamor	Arun	Dudhkoshi	Likhu	Tamakoshi	Sunkoshi	Indrawati
Mt. Kumbhakarna	Tibet	Dudhkunda	Mt. Rolwaling	Tibet	Tibet	Mt. Jugal
Papdhani	Maprabha		Hemganga			

Saptagandaki-

Trishuli	Budhigandaki	Daraundi	Marshyangdi	Maadi	Setigandaki	Kaligandai
Tibet	Mt Ganesh	Mt. Buddha	Mt. Annapurna	Mt. Annapurna II	Mt. Annapurna	Mustang
Dharmadhara	Yashodhara	Wishwadhara	Shitprabha	Ratnadhara	Shuwarnsarita	Krishna gandaki

Saptakarnali-

Sanobheri	Thulobheri	Mugu Karnali	Tila	Humla Karnali	Budhiganga	Seti Nadi
Mt. Dhaulagiri	Mt. Mukul	Mt. Laddakh		Tibet	Jagadulla	Mt. Saipal

Rivers	Place of Origin
Bagmati	Bagdwar
Kaligandaki	Mustang
Seti	Mt. Annapurna
Madi	Lamjung Mountain
Likhu	Rolwaling Mountain
Budhigandaki	Laddakh Mountain
Jhimruk	Gaumukhi
Humla Karnali	Tibet (Besides Mansarovar)
Marshyangdi	Damodar Mountain
Dudhkoshi	Mahalangur
Tamor	Kumbhakarna Mountain
Budhiganga	Lake Jagadulla
Mahakali	Mt. Api / Mt. Jaskar
Indrawati	Lake Panch Pokhari of Mt. Jugal
Tinau	Chure
Thulobheri	Mt. Kanjirowa
Bishnumati	Sapantirth
Daraudi	Buddha Mountain
Mechi	Lalikhark

Major lakes of Nepal

Tilicho Lake

Tilicho Lake is located at an altitude of 4,919 meters above sea level, nestled in the foothills of the Fakhche Himal. This scenic lake lies in the northern region of the Annapurna range, in the Manang district of Mustang. The lake is approximately 4 km long, 1.2 km wide, and about 200 meters deep. It is also known as Tiri-Cho or Tilijo. To the north of the lake lies Nilgiri, and to the south, the Annapurna Himal. The lake, surrounded by snow, water, and rocks, is incredibly picturesque, making it a highly attractive destination.

Rara Lake

Located in the Mugu district, Rara Lake is renowned as the largest lake in Nepal. It is 5.2 km long, 2.4 km

wide, and has a depth of 167 meters. The lake is situated at an altitude of 2,990 meters above sea level and is also referred to as Mahendra Tal.

Phoksundo Lake

Phoksundo Lake is located in the Dolpa district. Situated at an altitude of about 3,613 meters above sea level, the lake lies at the base of the Kanjirowa Himal in the southern foothills and near the head of the Jagdulla stream, a major tributary of the Bheri River. The lake appears long in the north-south direction and wide in the east-west direction. It is approximately 4.82 km long and 1.61 km wide. Locally, the lake is known as 'Rigmo.' The shape of the lake is triangular and elongated. It is the second-largest lake in the country, following Rara Lake, and the deepest lake in Nepal. Due to the extremely cold water, no aquatic life exists in the lake. The lake has an outlet in the form of the Suli Gad stream, which features a waterfall about 176 meters high.

Tsho Rolpa Lake

Tsho Rolpa Lake is located in the Dolakha district, at an altitude of 4,580 meters above sea level. The lake is approximately 3 km long, 0.5 km wide, and has a depth of 100 meters. It is estimated to contain around 80 million cubic meters of water. Due to the potential risk of an explosion, a siphon system has been installed to release water and prevent the lake from bursting.

Phewa Lake

Located in Pokhara, Kaski district, Phewa Lake is 4.8 km long, 1.5 km wide, and has a depth of 24 meters. The lake is particularly attractive as it reflects the shadow of Machapuchhre, making it not only beautiful but also an important tourist destination.

Rani Pokhari

Rani Pokhari was built by King Pratap Malla to console his queen Anantpriya, who was grieving the loss of their son. The pond is 180 meters long, 140 meters wide, and covers an area of nearly 32 ropanis. At the northeast and northwest corners of the pond, Bhairab is enshrined, while at the southeast corner, Mahalaxmi, and at the southwest corner, a statue of Ganesh is placed.

Toudah

Toudah, located in Kathmandu, is associated with a mythological story in which Manjushree is said to have cut through the Chobhar gorge with a sword or Lord Krishna used his discus to release the water from the Kathmandu Valley. After this, the Nagas were required to leave the area, which led to the creation of a pond at Chobhar Bhutkhel, where the king of the Nagas, Karkotak, was given a place to reside. Therefore, Toudah is also known as the residence of Karkotak Naga. It spans 96 ropanis of land, including an 84-ropari water area. With numerous species of birds found here, it is an attractive place for the study and research of birds.

Ghoda Ghodi Lake

Ghoda Ghodi Lake, located in the Kailali district, lies to the north of the Mahendra Highway, covering an area of 1.5 hectares in the shape of a horse's hoof. The lake has four or five small mounds (thumkis) in the center. The lake is named after the nearby Ghoda Ghodi Temple, which is located by the lake. The surrounding jungle is home to rare bird species like the hornbill, along with various other animals and birds. The lake also harbors fish, snakes, turtles, and other aquatic creatures.

Begnas Lake

Begnas Lake, located on the opposite slope of Pchabhaiya Hill, is situated at an elevation of 677 meters above sea level. This lake, which stretches across 225 hectares, has a depth of 7.5 meters. It lies close to Rupatal and is another prominent lake in the region.

Gosainkunda

Gosainkunda, a significant religious site, is located in the Rasuwa district of Bagmati Province. It is a popular pilgrimage destination, with thousands of devotees visiting on the day of Janai Purnima. The lake, situated at an altitude of approximately 4,360 meters above sea level, is known for its pristine water surrounded by

rough, black Himalayan rocks. The lake is formed by the confluence of the Sun Kunda, which is fed by waterfalls from the east and north.

Rupa Lake

Rupa Lake is located in the northeastern part of Pokhara, Kaski district, at the base of the Pchabhaiya mountain. The lake lies at an altitude of 701 meters above sea level, with a depth of 4.5 meters and spreads across an area of 120 hectares. Fishing activities are currently being carried out in the lake, providing an economic benefit to the local community.

Summary of Important Lakes in Nepal

Lake Name	Location District	Altitude (m)	Length (km)	Width (km)	Depth (m)
Tilicho Lake	Manang	4,919	4	1.2	200
Phoksundo Lake	Dolpa	3,613	4.8	1.61	650
Rara Lake	Mugu	3,200	5.2	2.4	167
Tsho Rolpa Lake	Dolakha	4,580	3	0.5	100
Phewa Lake	Kaski	900	4.8	1.5	24

Lakes and District of their location:

District	Lakes/Ponds
Kaski	Phewa, Rupa, Begnas, Dipang, Mairi, Khaste, Badhuwa, Syarkhu & Shanti Kunda
Rasuwa	Gosainkunda, Bhairabkunda, Saraswatikunda, Suryakunda, Parwatikunda, Ganeshkunda, Naagkunda & Dudhkunda
Dang	Bhakra Lake, Syanglang Lake, Bhote Lake, Range Lake, Saptakund, Jalkasanni Lake, Kakchhakani Lake
Palpa	Satyavati Lake, Nandan Lake, Suke Lake
Mugu	Rara Lake (Mahendra Lake), Rinmokscha Lake, Chhayana Lake, Papakund
Chitwan	Gadwa Lake, Kasara Lake, Bimorghaila Lake, Mujura Lake, Nandabhouja Lake, Beeshazar Lake, Sorahazar Lake
Myagdi	Bhale Basne Lake
Baglung	Rudra Lake, Jalpa Lake, Ganaunde Bhaifutta Lake
Dolpa	Putha Lake, Jagadulla Lake, Phoksundo Lake, Suke Lake, Kali Lake
Gorkha	Kalchhuman Lake, Kalpokhari, Runche Lake, Nag Kund
Rupandehi	Gajedi Lake
Parbat	Baraha Lake
Udayapur	Rauta Pokhari, Jogi Lake
Arghakhanchi	Thada Lake, Syalang Lake
Bhojpur	Kalpokhari, Salpa Pokhari
Jumla	Kagari Lake, Shey Lake, Kavi Lake
Kanchanpur	Beikote Lake, Khilikmala Lake, Banda Lake, Rani Lake
Kathmandu	Ranipokhari, Kamalpokhari, Panipokhari, Nagpokhari, Taudaha, Katuwal Lake
Sankhuwasabha	Sabha Pokhari, Siddha Pokhari, Matsya Pokhari, Panchpokhari, Dui Kund
Sindhupalchok	Panchpokhari, Baulaha Pokhari
Bardiya	Budaiya Lake
Sarlahi	Pattar Pokhari
Doti	Saptad Lake, Sun Lake
Bara	Kamini Lake
Lamjung	Deupokhari, Indrapokhari, Barahpokhari, Kanma Pokhari, Duipokhari, Ilampokhari
Panchthar	Jorpokhari, Kimbu Pokhari
Sunsari	Bulbule Lake, Samrajani Lake, Murali Lake, Sau Lake
Sindhuli	Gajuli Lake
Dadeldhura	Resunga Lake
Parsa	Banjari Lake
Dolakha	Jata Pokhari, Chhorolpa Glacier Lake

Forest Resources:

The topography, climate, soil, and other factors of a place influence its vegetation. Since climate affects vegetation, there is an interdependent relationship between the two. Forest resources hold a significant place in Nepal's environmental conservation and balance. According to data from 1954, more than 45% of the country's land area was covered by forests. However, by 1979, this figure dropped to 43%, further declining to 37.4% in 1986. A survey conducted in 1998 reported the forest area at only 39.6%, but the 2014 forest survey revealed an increase, with forest cover reaching 44.74%.

In the past, large areas were deforested for settlement and rehabilitation purposes, while some regions faced encroachment on forests. Based on Nepal's diverse topography and forests, the vegetation of Nepal can be classified into the following five groups:

(a) Sub-Tropical Evergreen Forest:

The sub-tropical evergreen forests found up to an altitude of 1,200 meters in the Terai, Bhabar, Dun, and Chure hills experience sufficient heat and rainfall, resulting in tall, thick, robust, and evergreen trees. Stretching parallel from east to west, this forest was once spread across four *kos* (about 12 kilometers), hence it is also called the *Charkose Jhadi*. Major vegetation in this region includes **sal**, **sissoo**, **khair**, and **silk cotton (simal)**, along with plants like **sawye**, **dhaddi**, and **hemp (ganja)**.

This forest is home to wildlife such as elephants, rhinoceroses, tigers, bears, and deer, making it economically and biologically significant.

(b) Temperate deciduous forests:

In Nepal, temperate deciduous forests are found at elevations ranging from 1,200 meters to 2,100 meters, particularly in the upper parts of the Chure hills, the midsection, and the lower parts of the Mahabharat range. The trees in these forests are tall and thick. While the lower parts are dense and shaded, the upper regions are colder and experience frost, causing most leaves to shed, which is why this forest type is referred to as "deciduous." In the higher elevations, there is a mix of deciduous and coniferous trees. Among the evergreen, deciduous, and coniferous tree species found here, notable ones include Sal, Pine, Deodar, Champ, Katus, Walnut, Rhododendron, Peepal, Chilaune, Banyan, Bamboo, Silk Cotton Tree (Simal), Uttis, Palash, Lakuri, and Champ. This type of forest is considered of lower quality compared to tropical evergreen forests.

(c) Temperate Evergreen Coniferous Forests:

Coniferous forests are found at elevations ranging from 2,101 meters to 3,350 meters in the upper parts of the Mahabharat range and the lower parts of the Himalayan mountains. Due to the year-round cold climate and frequent snowfall, the trees and their leaves are pointed, resembling cones. The soil in these areas is porous, supporting evergreen forests. Medium-sized softwood trees are commonly found here, with notable species including Champ, Pine, Deodar, Katus, Dhupi, Simal, Chilaune, Cane Bamboo (Betbans), Uttis, Bamboo, Rhododendron, and Bhojpatra.

(d) Alpine Vegetation or Grasslands:

Alpine vegetation refers to the plant life found in the hilly and lower Himalayan regions of Nepal, at elevations ranging from 3,351 meters to 5,000 meters, where temperatures are low and rainfall is minimal. Due to the cold and dry climate at this altitude, tall, thick, and hardwood trees cannot thrive. Among alpine vegetation, low-altitude rhododendrons, dwarf bamboo (Nigale), and some shrub-like trees are found up to an elevation of 3,600 meters, while vast grasslands are found above this altitude. The higher regions are home to colorful wildflowers and various medicinal herbs.

(e) Vegetation of Cold Desert/Tundra Vegetation:

In Nepal, at elevations above 5,000 meters, snowfall prevents vegetation from thriving. As a result, only mosses and lichens are found in steep areas where snow does not accumulate. The lichens and mosses that manage to grow in such inhospitable conditions are referred to as tundra vegetation or cold desert vegetation.

According to the Forest Act, 2049, the classification of forests is as follows:

1. National Forest
2. Government-Managed Forest
3. Community Forest
4. Leasehold Forest
5. Religious Forest
6. Private Forest

National Parks of Nepal

S.N.	National Park	Established (B.S.)	Location	Area (sq. km)	Main Features
1	Chitwan National Park	2030	Chitwan, Makwanpur, Parsa, Nawalparasi	952.63	One-horned rhinoceros, python. Area expanded on 2073/7/1.
2	Sagarmatha National Park	2032	Solukhumbu	1148	Musk deer, Himalayan bear
3	Langtang National Park	2032	Rasuwa, Nuwakot, Sindhupalchok	1710	Musk deer, Himalayan wildlife
4	Rara National Park	2032	Mugu, Jumla	106	Home to Rara Lake and Himalayan flora/fauna
5	Bardiya National Park	2032	Bardiya	968	Swamp deer (Barasingha), blackbuck
6	Shey Phoksundo National Park	2040	Dolpa, Mugu	3555	Tibetan hare, blue sheep
7	Khaptad National Park	2042	Bajhang, Bajura, Doti, Achham	225	Wild dogs, leopard cat, Dhorai
8	Makalu-Barun National Park	2048	Sankhuwasabha, Solukhumbu	1500	Red panda, Olive Ground thrush
9	Shivapuri-Nagarjun National Park	2058	Kathmandu, Nuwakot, Sindhupalchok, Dhading	159	Assamese monkey, pangolin
10	Banke National Park	2067	Banke, Salyan, Dang	550	Tiger, four-horned antelope
11	Shuklaphanta National Park	2031	Kanchanpur	305	Became a national park on 2073/11/19.
12	Parsa National Park	2040	Chitwan, Makwanpur, Bara, Parsa	627.39	Declared a national park on 2074/2/19.

Conservation Areas of Nepal:

S.N.	Conservation Area	Established Year (B.S.)	Area (sq. km)	Geographical Location	Notable Wildlife and Features	Gazette Notification Details
1	Annapurna	2049	7,629	Lamjung, Manang, Mustang, Myagdi, Kaski	Rare Himalayan wildlife, vegetation, and cultural heritage	Declared on 2049/04/05
2	Kanchenjunga	2054	2,035	Taplejung	Snow leopard, musk deer, Himalayan brown bear, wolf, goral	Declared on 2054/04/06
3	Manaslu	2055	1,663	Gorkha	Snow leopard, musk deer, Himalayan tahr, various bird species	Declared on 2055/09/13

4	Krishnasar	2065	16.95	Bardiya	Blackbuck (Krishnasar), tiger, python	Declared on 2065/12/03
5	Gaurishankar	2066	2,179	Ramechhap, Dolakha, Sindhupalchok	Snow leopard, musk deer, red panda, pheasants like Danphe, Munal	Declared on 2066/09/27
6	Api Nampa	2067	1,903	Darchula	Snow leopard, musk deer, black bear, Himalayan thar, medicinal herbs like Yarsagumba	Declared on 2067/03/28

Minerals in Nepal

The elements, compounds, and substances found within the Earth's crust are known as minerals. Minerals are extremely useful for economic development and prosperity. There are many examples of countries that have become prosperous due to abundant mineral resources. If minerals can be mined and utilized effectively, and industries based on these minerals can be established, there is a significant possibility of earning a substantial amount of foreign currency.

Below is a list of some major minerals found in Nepal and their potential locations.

Mineral	Locations
Iron	Phulchoki, Thos, Chitwan, Kulekhani, Bhainse, Pyuthan, Bajhang
Copper	Kulekhani, Buddhakhola, Myagdi, Dhankuta, Bahrabise, Bandipur, Chisapani
Mica	Sundarjal, Bhojpur, Lamjung, Biang, Gosainkunda
Lead	Phulchoki, Baglung, Rasuwa, Mt. Ganesh, Baitadi, Tiptang
Gold	Kaligandaki, Budhigandaki, Sunkoshi
Limestone	Godawari, Bhainsedobhan, Chobhar, Hetauda, Palpa, Jogimara
Silver	Baglung, Chisapanigadhi, Phulchoki
Gas	Terai side, Dailekh, Mustang, Kathmandu Valley
Petroleum	Koilabas of Dang, Nepalgunj, Dhangadhi, Muktinath, Pyuthan, Dailekh
Sulphur	Goshainkunda, Chisapanigadhi, Barah area
Magnesite	Mt. Ganesh, Dolakha, Udaypur
Khari Dhunga	Sindhupalchok, Khotang, Bhojpur
Slate	Bandipur, Daraun of Syanja, Bagamati Lumbini, Janakpur
Coal	Chitwan, Dang, Salyan, Chatara, Kathmandu
Geru	Chautara-Sindhupalchok
Cobalt	Dhankuta, Palpa, Gulmi, Arghakhachi
Marble	Godawari, Mahabharat range
Soda	Doti, Salyan
Black Salt	Rasuwa, Syaprubenshi
Nickel	Khokling, Nampa, Bhorle
Salt	Thakkhola-Mustang
Graphite	Sundhupalchok, Palpa

Ramsar Convention

The Ramsar Convention is an agreement made by environmental conservationists worldwide for the protection of ecologically significant wetlands. It was signed on February 2, 1971, in Ramsar, Iran. This convention came into force on December 21, 1975. It commits to the protection of major wetland areas around the world. Based on the data from May 2018, the Ramsar Convention has recognized 2,331 wetlands globally as being worthy of protection from an ecological perspective. Nepal has been a contracting party to this convention since 1987. Under this convention, 10 wetland areas in Nepal are currently listed as international wetlands.

Wetland Areas of Nepal Listed as Ramsar Sites:

S.No.	Name	Listed Year	Location	Area (sq. km)
1	Kashi Tappu Wildlife Reserve	1988.12.17	Sunsari	175
2	Besh Hazari Tal	2003.08.13	Chitwan	32
3	Ghoda Ghodi Tal	2003.08.13	Kailali	25.63
4	Jagdishpur Tal	2003.08.13	Kapilvastu	2.25
5	Sagarmatha National Park	2007.09.23	Solukhumbu	77.70
6	Shey Phoksundo Area	2007.09.23	Dolpa	4.94
7	Rara Tal and Park	2007.09.23	Mugu	15.83
8	Gosainkunda	2007.09.23	Rasuwa	10.13
9	Mai Pokhari	2008.11.27	Ilam	0.9 (smallest)
10	Phewa Tal and Nine Other Lakes	2016.12.02	Kaski	261.06

The division of Nepal on the basis of climate and its impact on people's lives:

The climate of Nepal can be divided into five types of climate based on geographical structure and altitude.

(a) Sub-Tropical Monsoon Climate

This type of climate, known as semi-tropical and subtropical, is found in Nepal's Terai, inner Terai (Bhabar), Doon, and Chure regions at elevations up to 1,200 meters (4,000 feet). In the summer, the temperature can reach between 38 to 42 degrees Celsius, while in the winter it drops to between 15 to 5 degrees Celsius. During the monsoon season, the southwest monsoon winds coming from the Bay of Bengal bring rainfall. The rainfall generally decreases from east to west and increases from north to south, making the western Terai regions hotter than the eastern Terai. Specifically, regions like Bhairahawa, Nepalgunj, and Kailali experience extreme heat. Similarly, in the inner Terai areas like Dang and Surkhet, the heat is also quite intense.

(b) Warm Temperate Climate

This type of climate, known as warm temperate, is found in the Chure Hills and Mahabharat Mountain at elevations between 1,200 and 2,100 meters (4,000–7,000 feet). In this climate, summers are warm, while winters are quite cold. During the summer, the temperature ranges from 24 to 31 degrees Celsius, and in the winter, it can drop to 0 degrees Celsius. The monsoon winds, which bring rain, flow from the southern part, resulting in up to 250 centimeters of rainfall in the southern slopes, while the northern slopes receive up to 100 centimeters of rain.

(c) Cold Temperate Climate

This type of climate, known as cold temperate, is found in the Mahabharat Mountain range at elevations between 2,100 and 3,350 meters (7,000–11,000 feet). In this climate, summers are mild, while winters are very cold. The temperature during summer ranges from 15 to 20 degrees Celsius, and in winter, it can drop to 0 degrees Celsius. The region receives up to 100 centimeters of rainfall annually, and snow falls during the winter. Due to the low temperatures and low rainfall, agricultural activities are extremely limited here. While life is difficult, this climate is considered favorable for health.

(d) Alpine Climate

In Nepal, the alpine climate is found in the Himalayan region at elevations ranging from 3,350 to 5,000 meters (11,000–16,000 feet) above sea level. During the months of Chait (March–April), Baisakh (April–May), and Jestha (May–June), the temperature reaches up to 10°C, while for the remaining 9 months, the temperature remains below 0°C. The area receives up to 30 millimeters of rainfall. Agriculture is not possible here, but due to the large alpine pastures, the region is suitable for livestock grazing and tourism activities.

(e) Polar or Tundra Climate

The extremely cold and dry climate of the Himalayan region is referred to as polar or tundra climate. Snowfall is constant in this area, and precipitation occurs in the form of snow. The temperature in areas above 5,000 meters (16,000 feet) drops below zero.